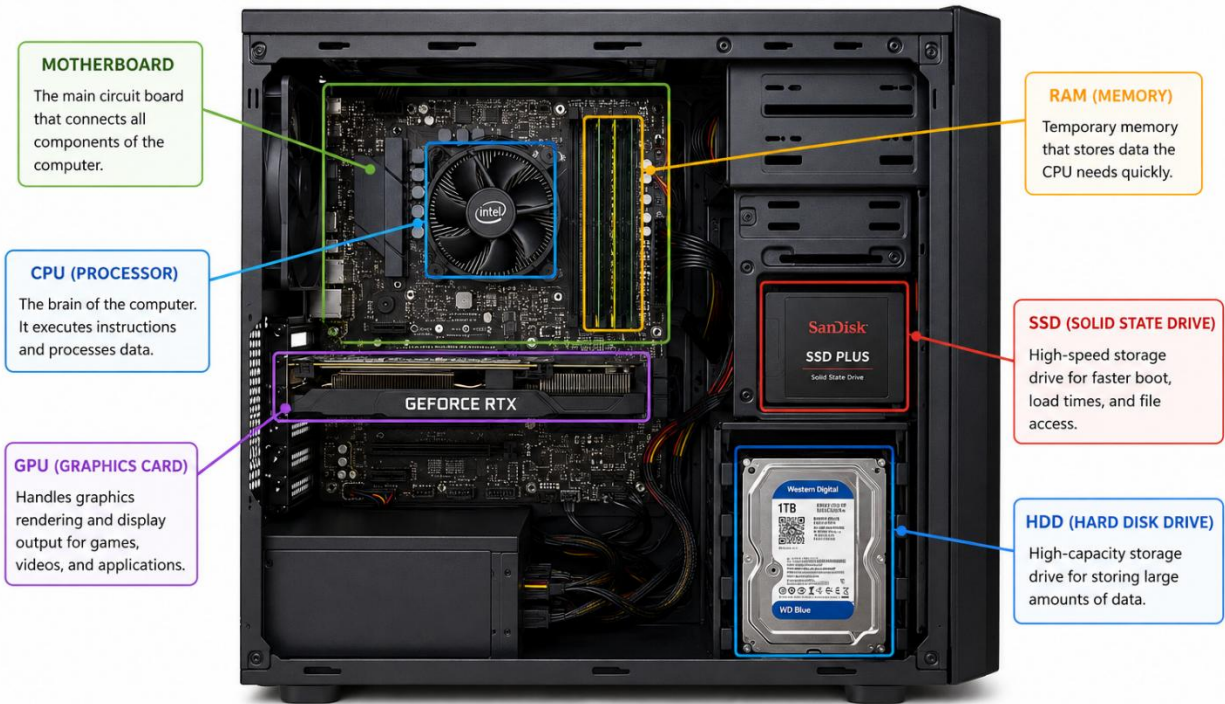


Assignment - 1.1

1. Draw a labeled diagram of a desktop PC, clearly marking the motherboard, CPU, RAM, GPU, and both HDD and SSD storage components.



2. Create a comparison table showing the main differences between HDD and SSD storage in terms of speed, durability, and typical use cases.



Feature	HDD (Hard Disk Drive)	SSD (Solid State Drive)
Storage Technology	Uses spinning magnetic disks (platters)	Uses flash memory (chips)
Speed	Slower read/write speed (80–160 MB/s)	Much faster read/write speed (500 MB/s to 7000+ MB/s)
Boot Time	Takes longer to start the computer	Boots the computer much faster
Durability	Less durable due to moving parts	More durable because it has no moving parts
Shock Resistance	Easily damaged by drops or shocks	Highly shock-resistant
Noise	Makes noise while operating	Silent operation
Power Consumption	Consumes more power	Consumes less power
Heat Generation	Produces more heat	Produces less heat
Cost	Cheaper per GB of storage	More expensive per GB of storage
Typical Use Cases	Desktop PCs, backup storage, file servers, budget computers	Laptops, gaming PCs, high-performance desktops,

HDD (Hard Disk Drive):

- Slower but less expensive.
- Best for storing large amounts of data such as videos, photos, and backups.

SSD (Solid State Drive):

- Much faster and more reliable.

- Best for installing the operating system, games, and applications that require high speed.

3. Write a short explanation (2-3 sentences each) describing the role of the CPU, RAM, and GPU in running a game like PUBG or Free Fire on a PC.



- **CPU (Central Processing Unit)**

The CPU is the brain of the computer. It processes game instructions, controls player movements, manages AI, and handles game physics. A faster CPU helps the game run more smoothly with better performance.

- **RAM (Random Access Memory)**

RAM temporarily stores the game's data while it is running. It allows the CPU to quickly access important information, reducing loading times and preventing lag. More RAM helps the game run smoothly, especially when multiple applications are open.

- **GPU (Graphics Processing Unit)**

The GPU is responsible for rendering the game's graphics, including characters, maps, lighting, and visual effects. A powerful GPU provides higher frame rates, better image quality, and a smoother gaming experience in games like PUBG and Free Fire.

4. Imagine you want to upgrade your PC to run the latest version of FIFA smoothly. List which hardware components you would consider upgrading and explain why for each.

- To run the latest version of **FIFA** smoothly on a PC, I would consider upgrading the following hardware

Hardware Component	Why Upgrade It?
Graphics Card (GPU)	A better GPU improves graphics quality, increases frame rates (FPS), and provides smooth gameplay with realistic visuals.
Processor (CPU)	A faster CPU processes game data, player movements, AI, and physics more efficiently, reducing lag and improving overall performance.
RAM	Increasing RAM (at least 16 GB) allows the game to load faster and run smoothly, especially when other applications are open.
SSD (Solid State Drive)	Replacing an HDD with an SSD reduces game loading times and speeds up system boot and game installation.
Power Supply (PSU)	A higher-capacity PSU may be required to safely support upgraded components such as a powerful GPU.
Cooling System	Better cooling prevents the CPU and GPU from overheating during long gaming sessions, maintaining stable performance.